MASTER'S THESIS DEFENSE · UNIVERSITY OF TRIESTE

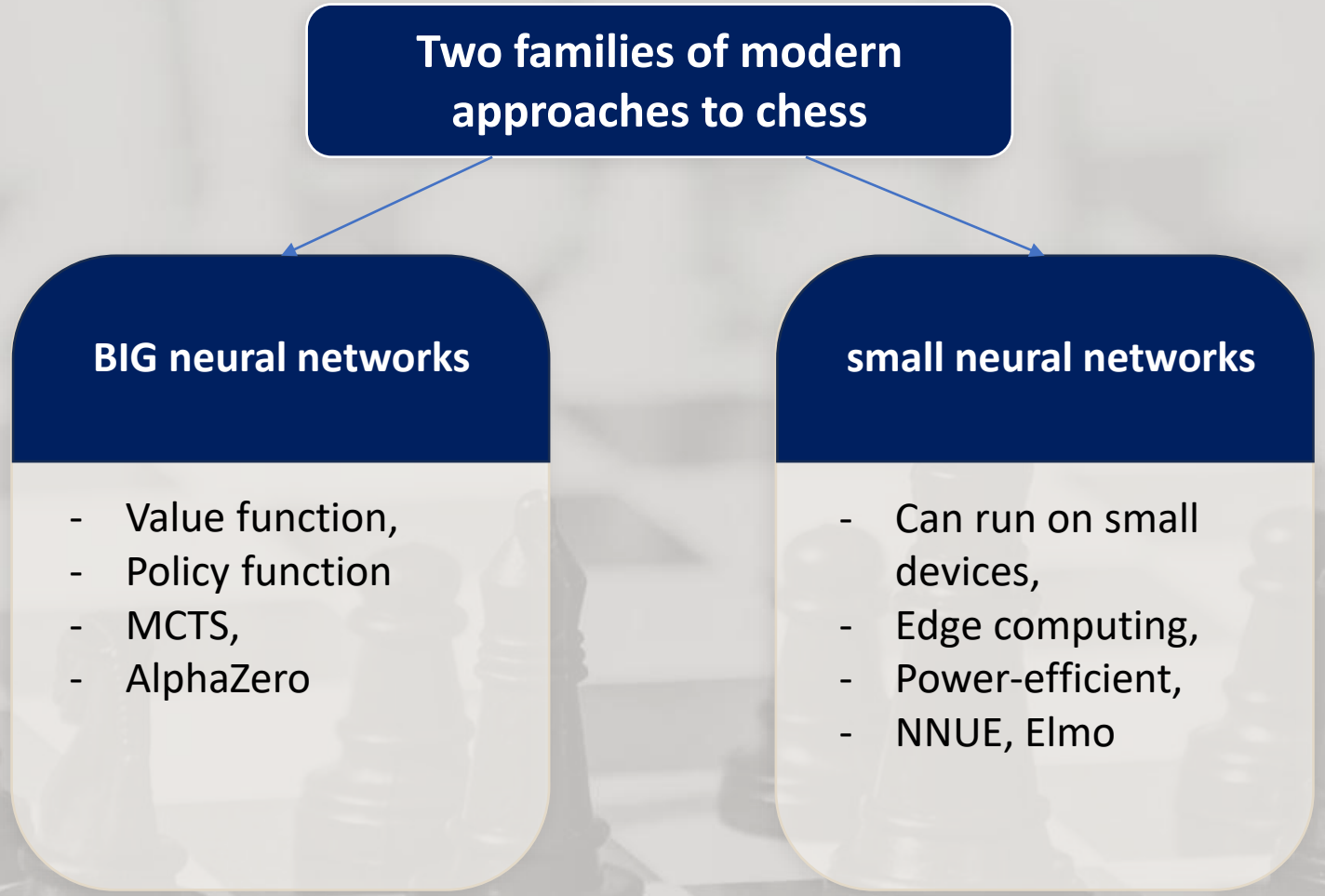
Unsupervised State Bucketing for Mixture-of-Experts in Resource-Constrained Chess Engines

Can a tiny chess engine choose the right expert?

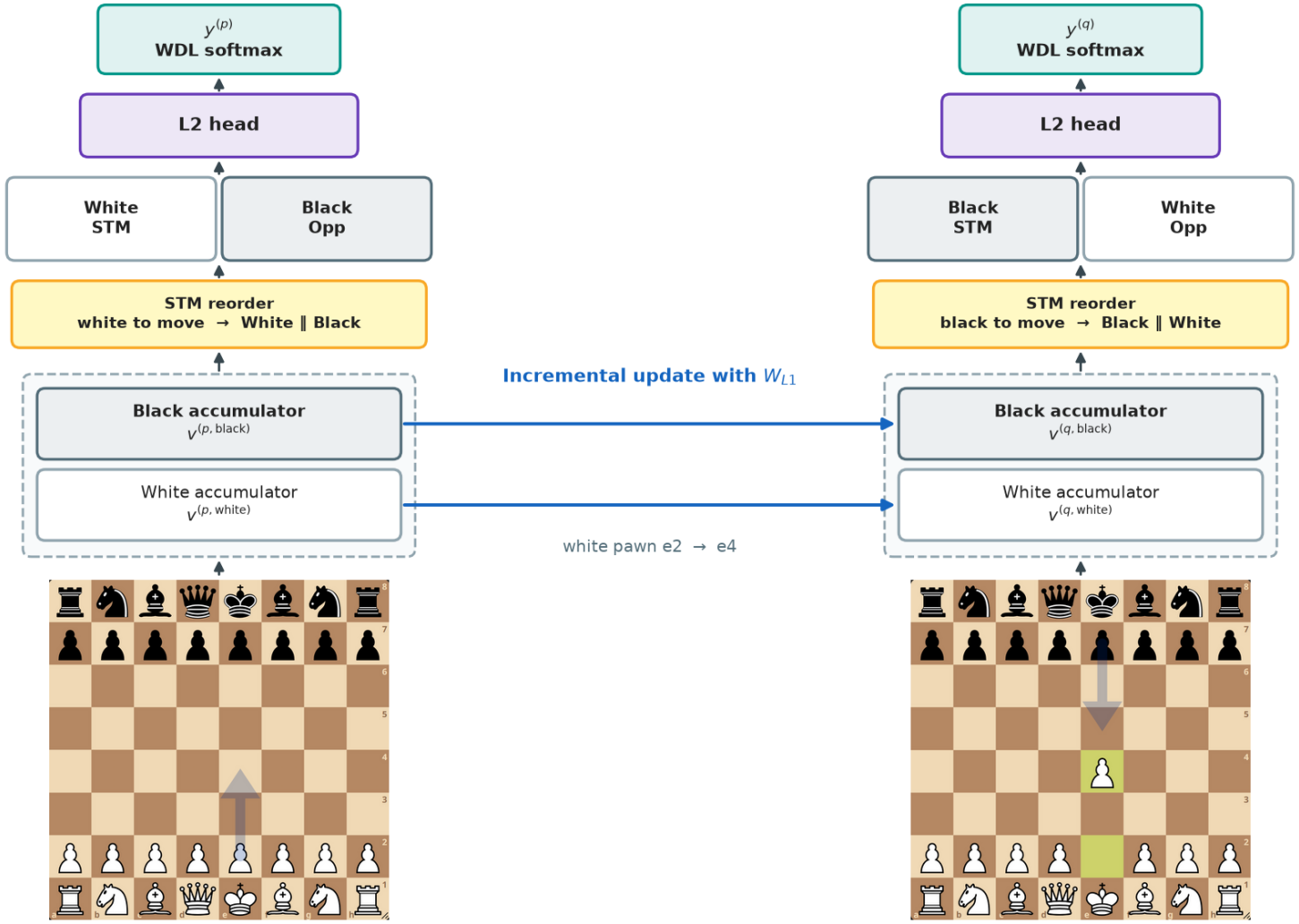
Omar Cusma Fait

Data Science and Artificial Intelligence
Thesis Supervisor: Alessio Ansuini, Area Science Park · Internship Supervisor: Marco Zennaro, ICTP

The Big Picture: AI on the Edge



The Base Architecture of the NNUE



Position p (white's turn)

Position q (black's turn)

NNUE updates only the changed features

1

A move touches few inputs

Of the 844 board features, only the squares that change are active.

2

Edit the accumulator

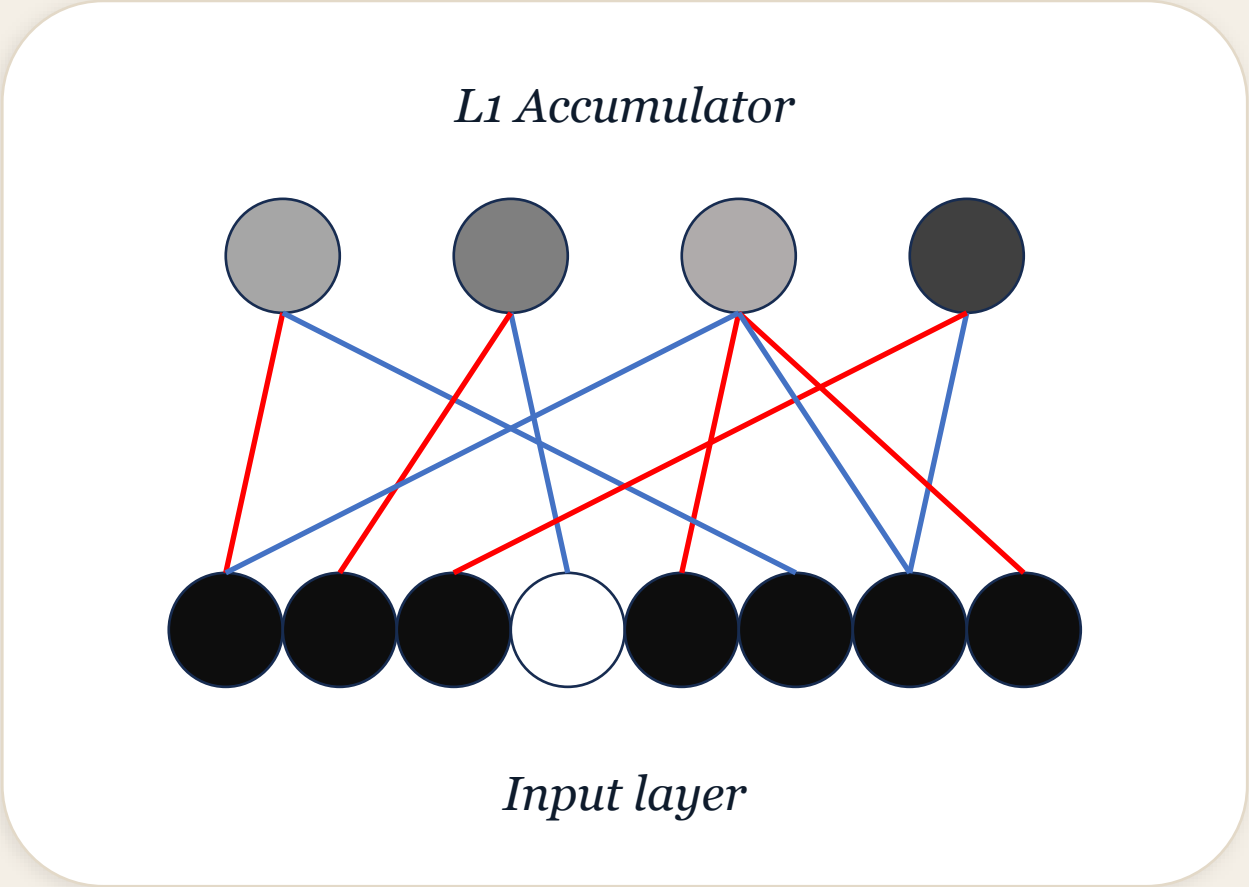
Subtract the removed features and add the new ones. The rest stays.

3

Reuse it on the next node

The L1 accumulator persists across moves, so evaluation stays cheap.

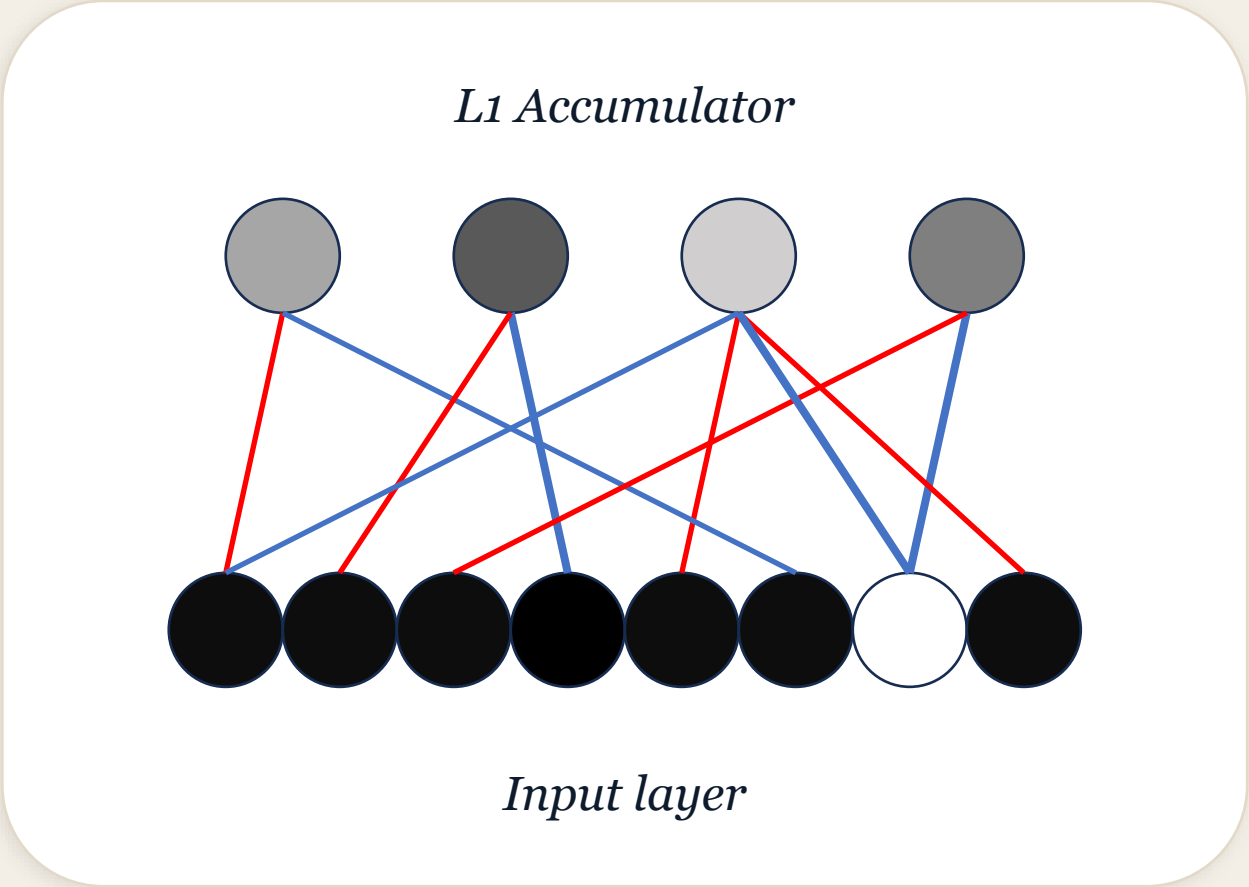
The Accumulator Layer



-

L1 is the shared first-layer representation. During inference, it’s activations are mostly recycled.

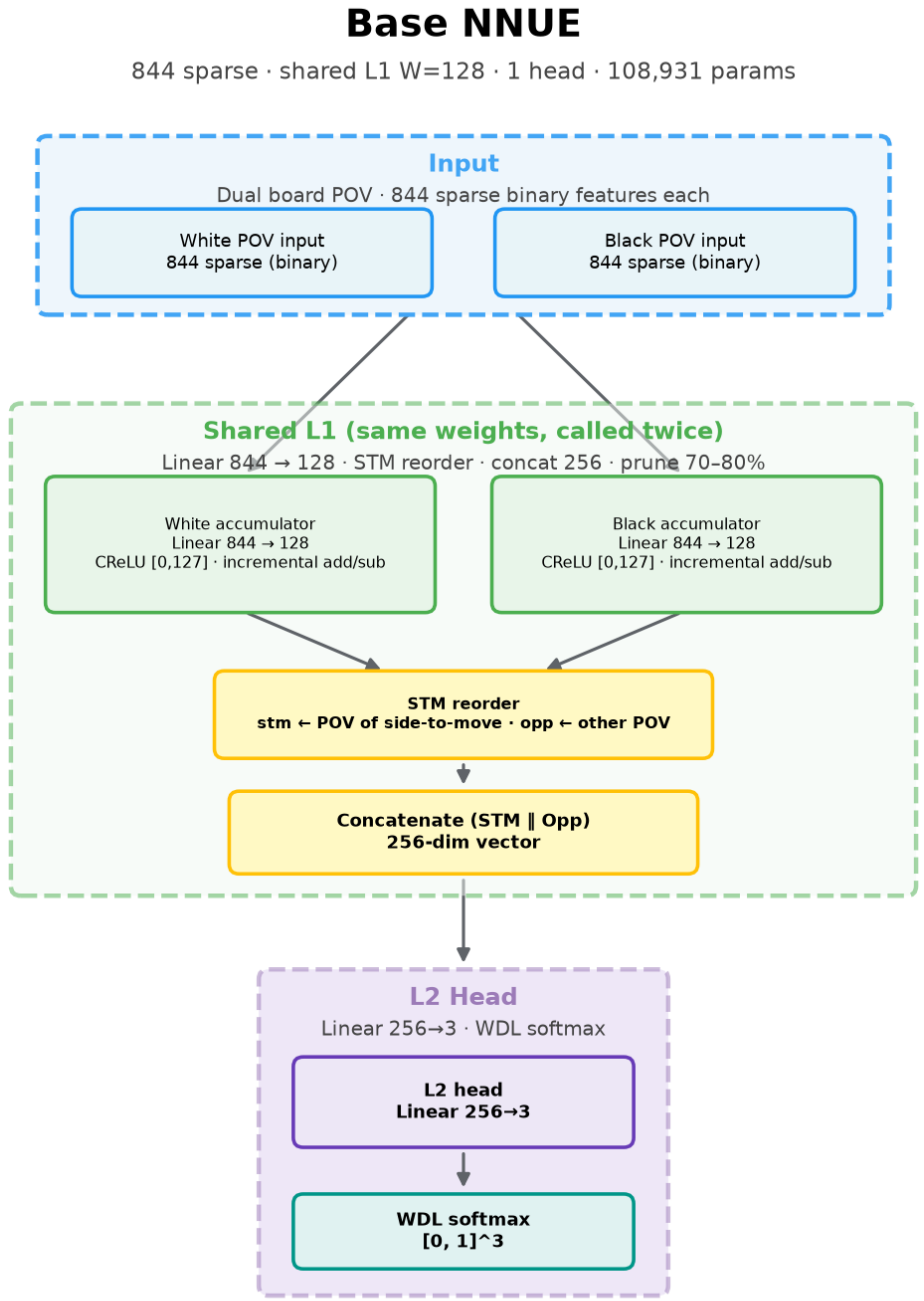
The Accumulator Layer



- L1 is the shared first-layer representation. During inference, it’s activations are mostly recycled.

The Architecture of the base NNUE

- Input layer
- Accumulator layer (L1)
- L2 hidden layer
- Output layer



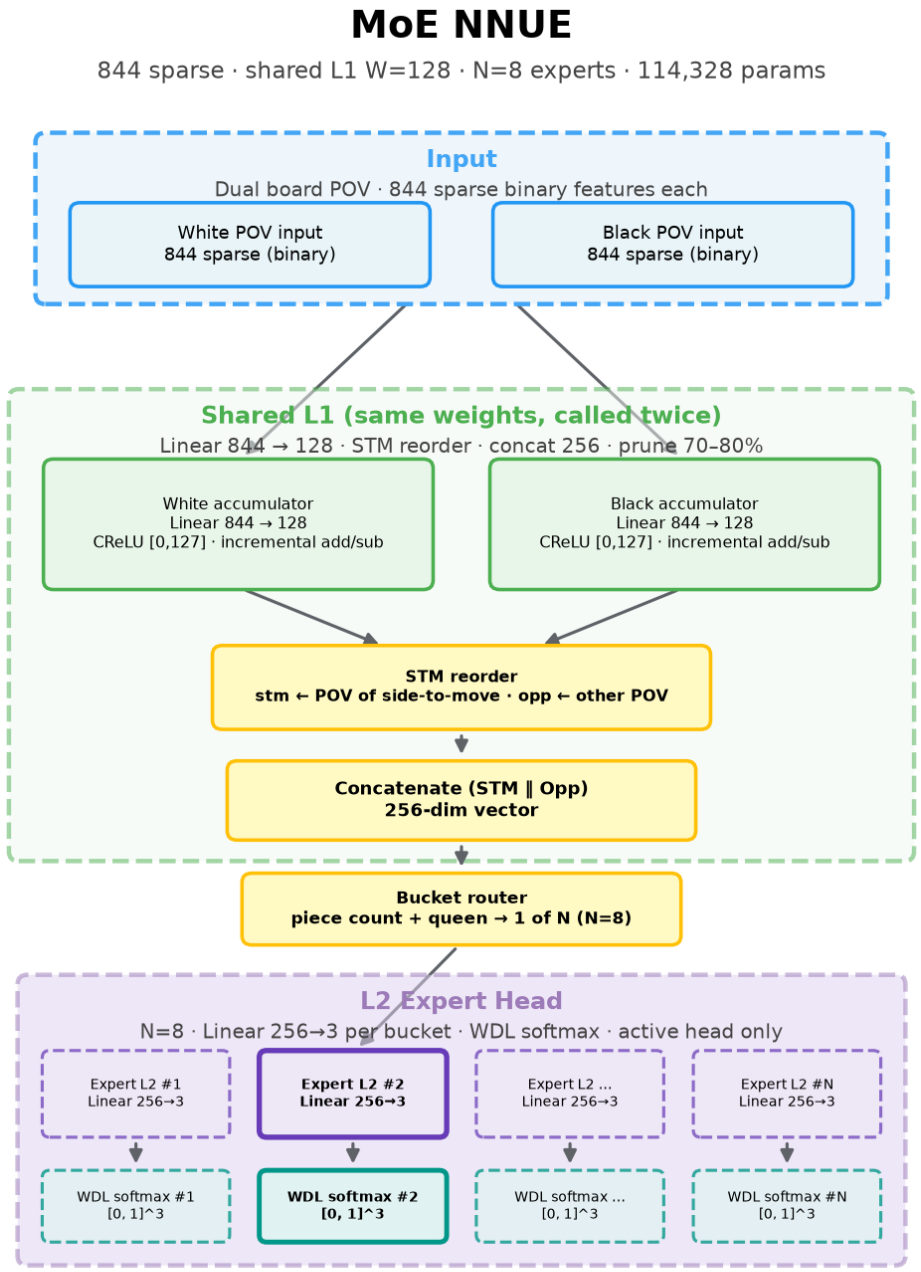
Student: DualHidden WDL · L1 W=64 · L2=128 · CE vs Lc0 STM (W,D,L)

encode_dual()

MoE NNUE ARCHITECTURE

The Architecture of the MoE NNUE

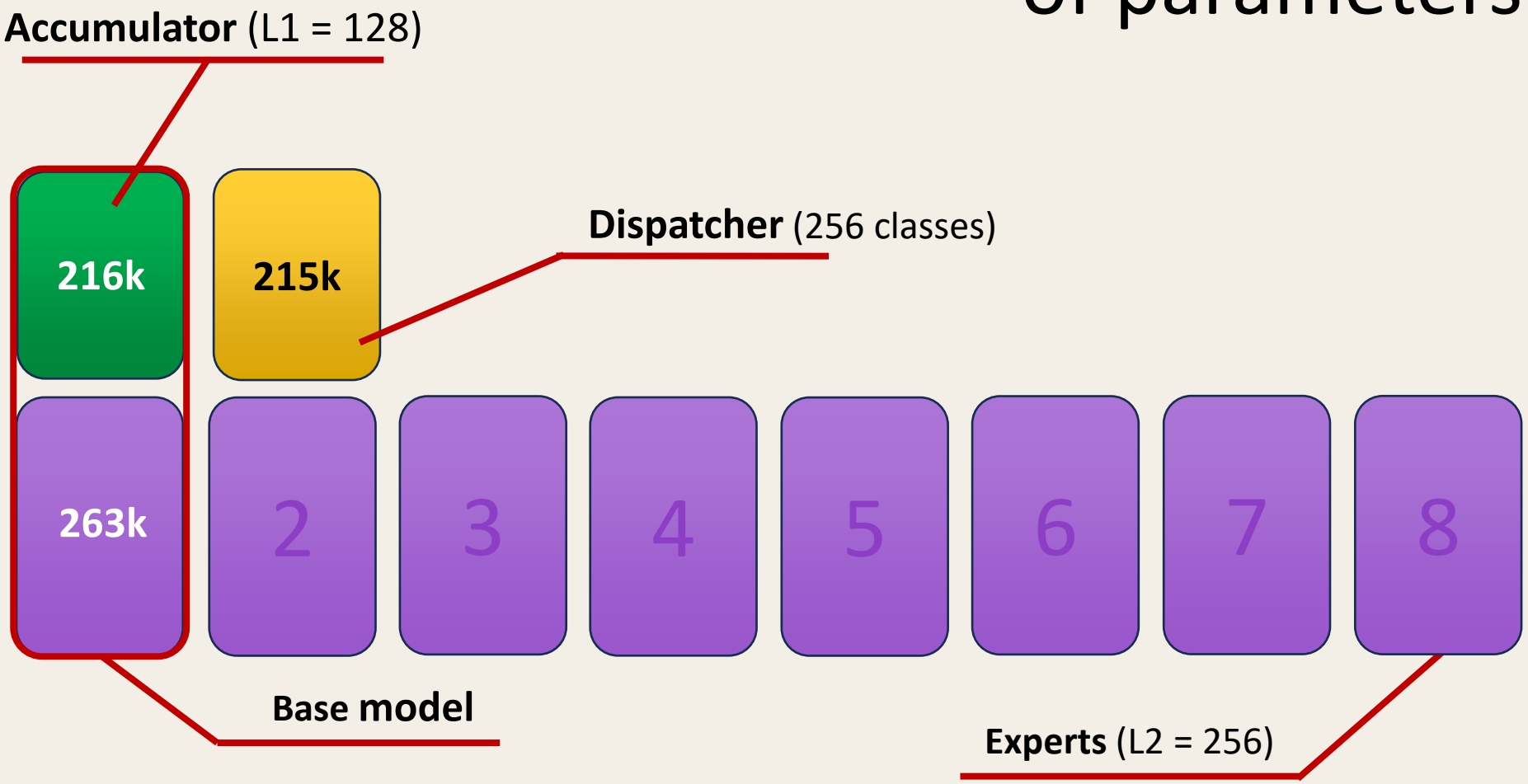
- A **router decides which head should be called**
- Expert heads can occupy a lot of memory
- During inference, only one head called (also the dispatcher)



Student: DualHidden WDL · L1 W=64 · L2=128 · CE vs Lc0 STM (W,D,L)

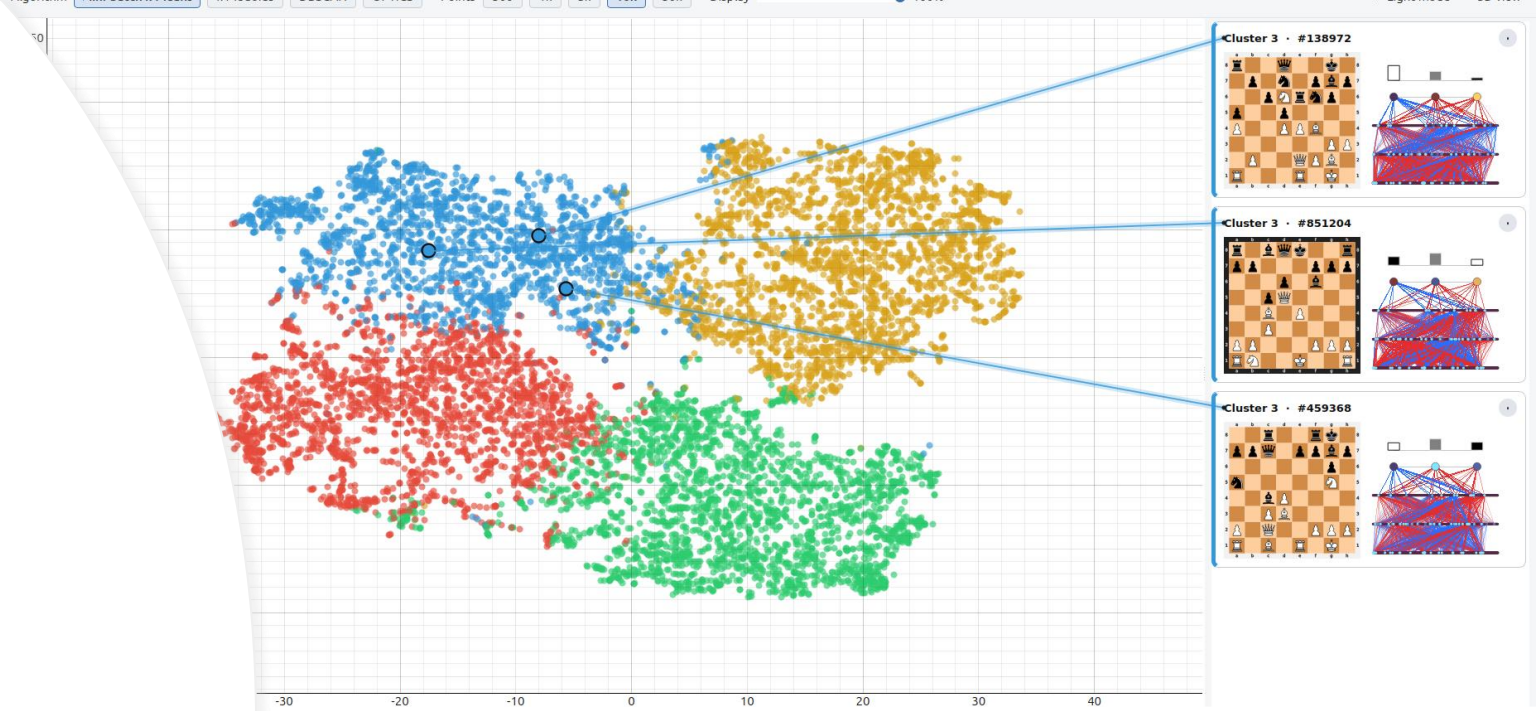
encode_dual()

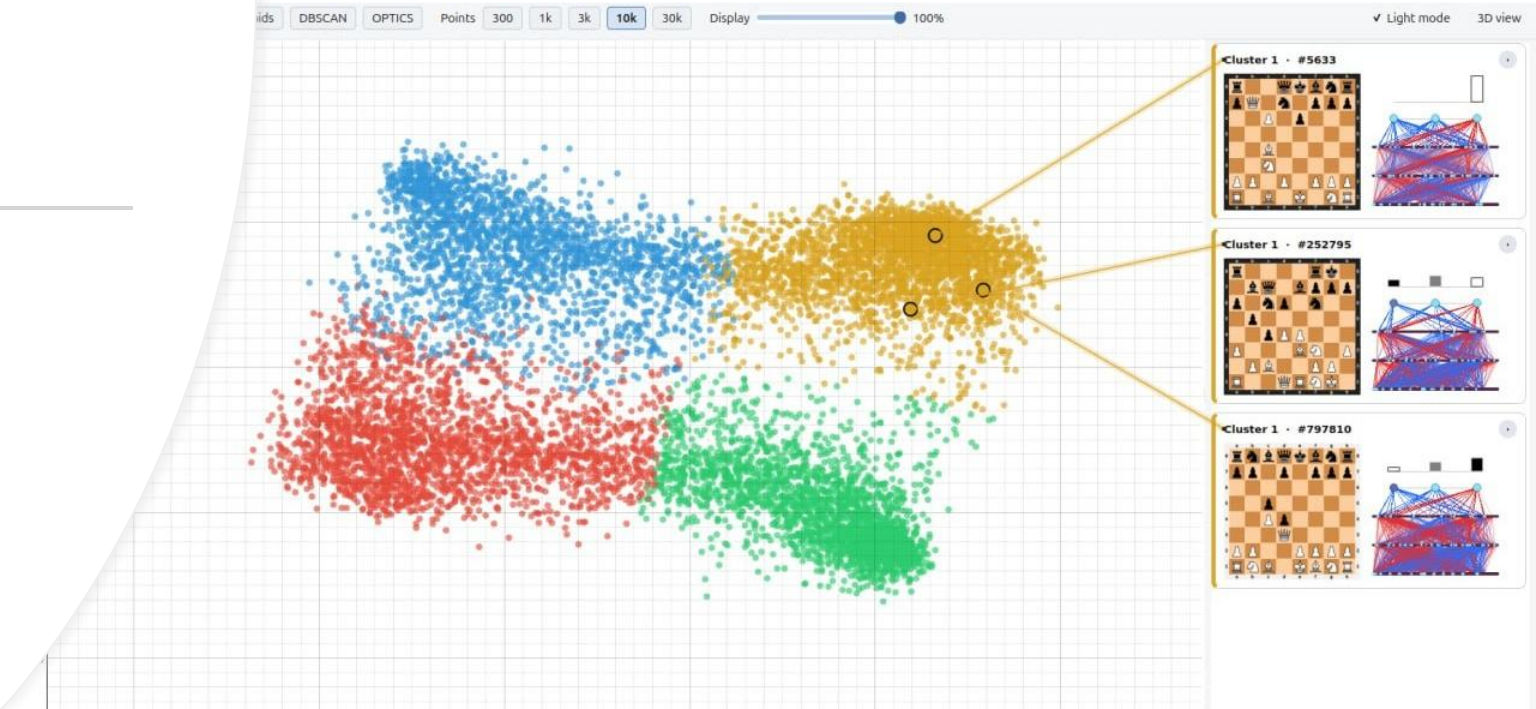
Number of parameters

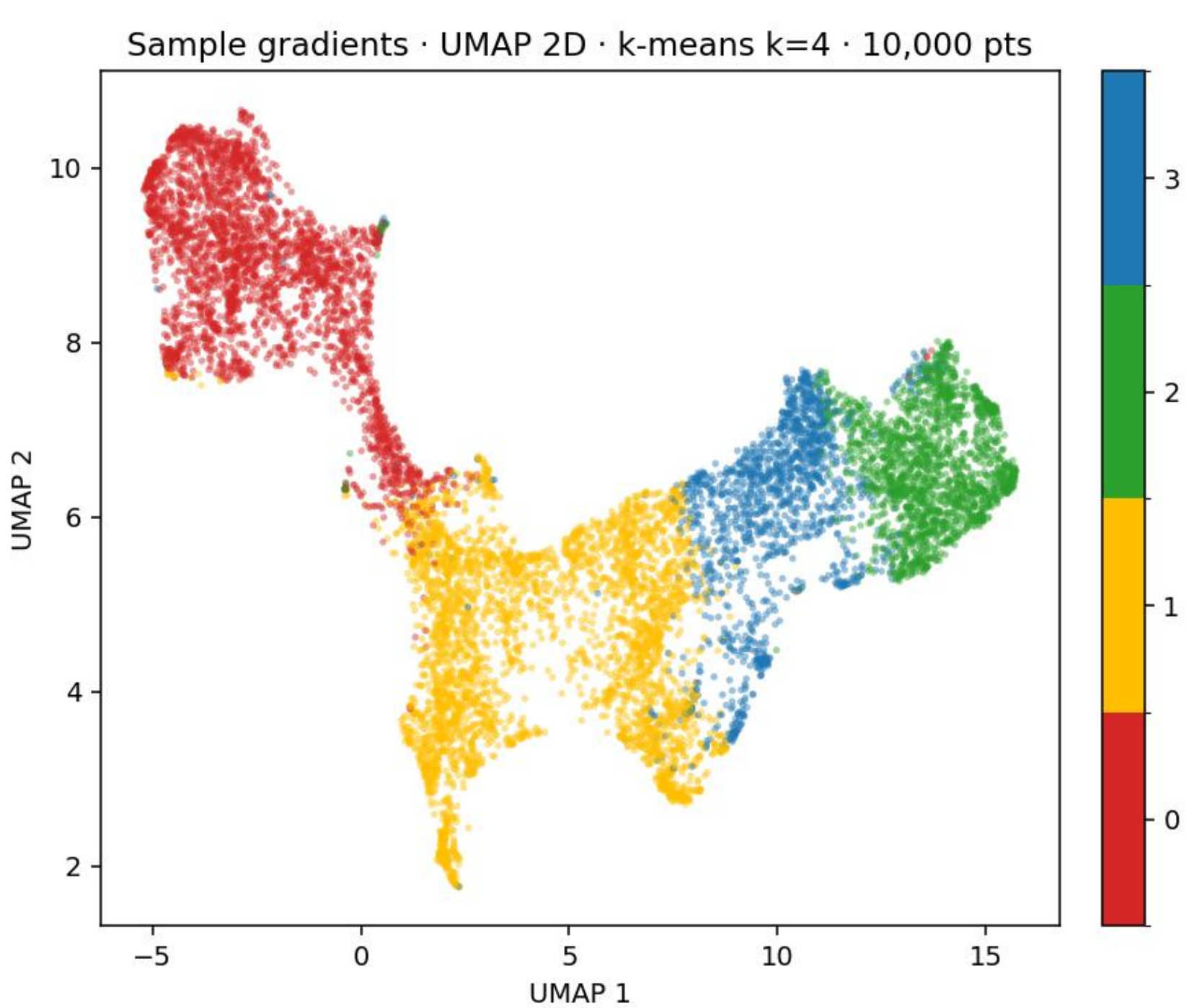


Representation of the number of parameters of the MoE NNUE

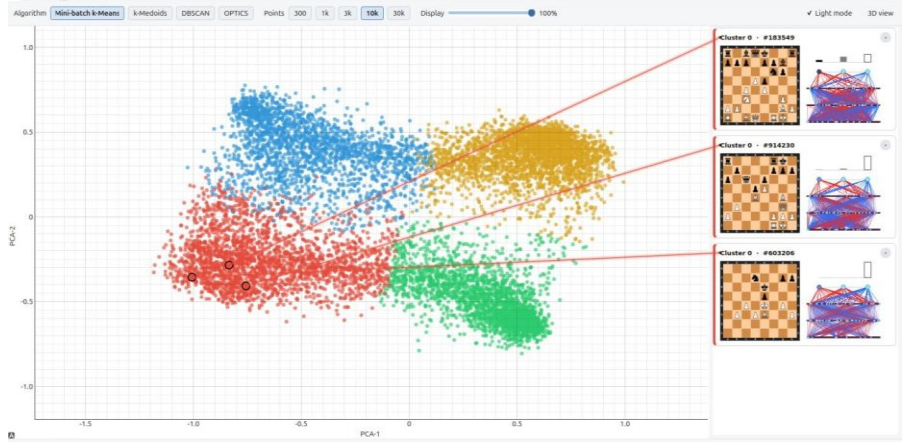
Clustering Sample Gradients













Clustering of the sample gradients

Training and evaluation protocol

150M

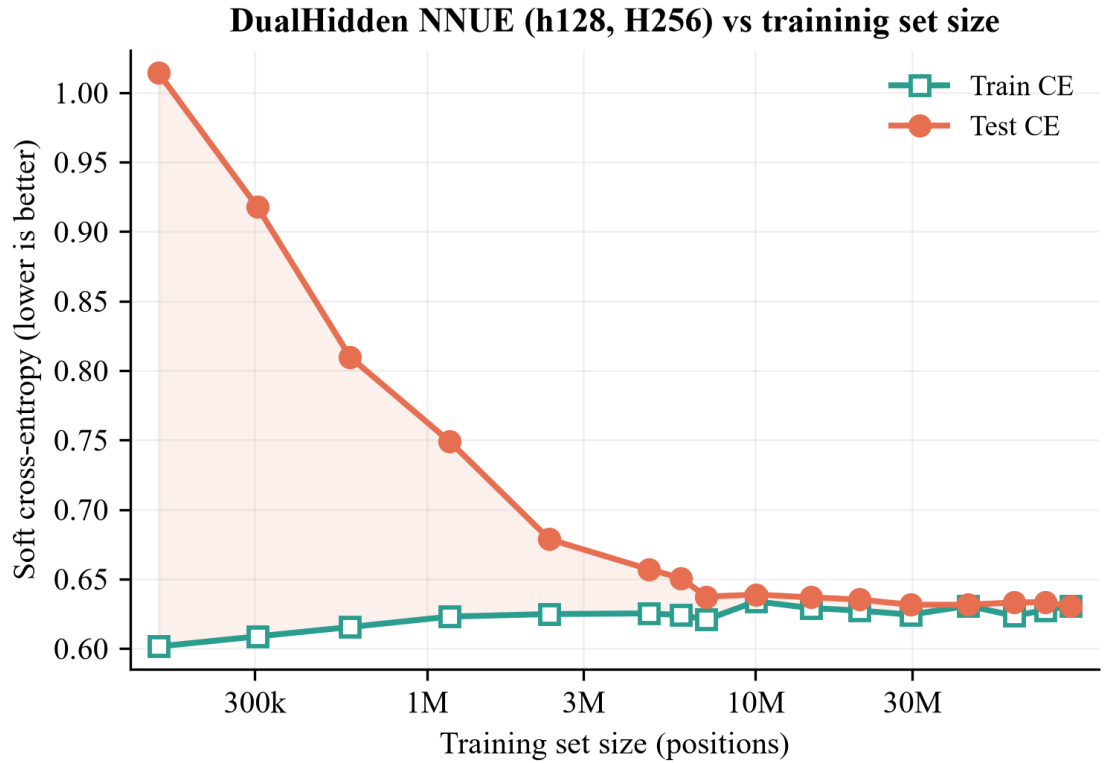
Training positions, mainly from Lichess.

~7M

Positions to overcome **overfitting**.

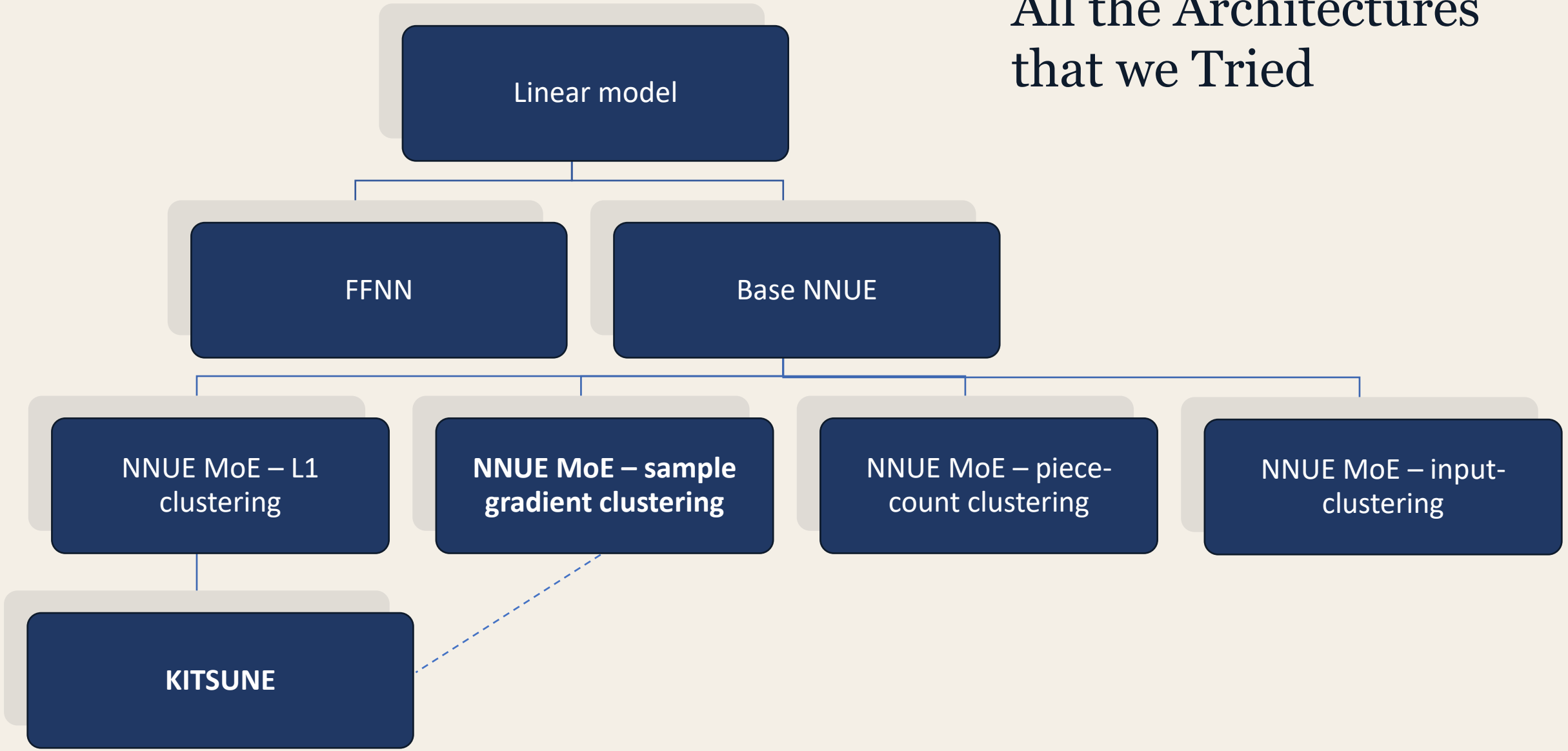
175k

Parameters in the biggest NNUE.



Example of overfitting due to lack of data

All the Architectures that we Tried



A diagram showing the implemented algorithms. The graph structure reflects the model architecture.

SAME L1 INPUT, SAME MLP DISPATCHER, EIGHT BUCKETS

Gradient buckets are hard to recover



45.5 percentage-point gap

A training signal that depends on the true value is difficult to infer from L1 alone.

Strong-base MoE gains are marginal

Model	K	CE
Reference NNUE	—	**0.6303**
Piece-count MoE	8	**0.6303**
L1-clustered hard MoE	16	**0.6300**
Gradient-clustered hard MoE	8	**0.6300**
Soft-gated MoE, top-2	2	**0.6304**

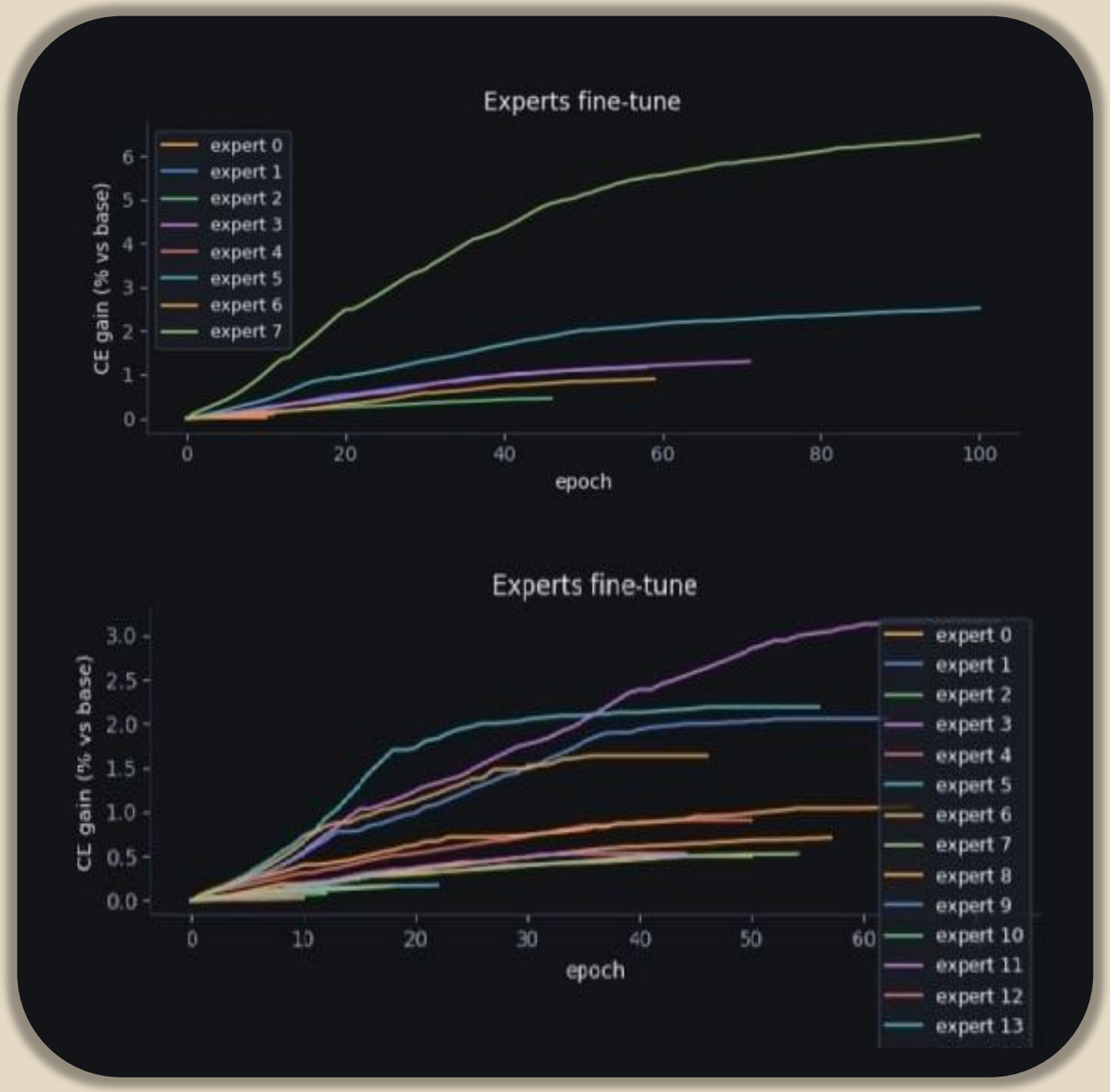
Best observed change in CE.

No decisive improvement over the reference NNUE… but why??

Tables 5.10–5.11, 50,000 held-out positions. Lower CE is better. These figures are not an Elo gain.

Expert fine-tuning step

- Some partitions produce **very effective experts**
- Many partitions produce **modest improvements**
- The depth of the **loss basen** is rougly proportional to the initial magnitude of the **gradient**



Some training trajectories for L1 clustering MoE NNUE

KITSUNE stitches micro-buckets offline

1

Partition L1

Cut the activation space into many small, convex micro-buckets.

2

Stitch offline

Group those buckets by their mean head gradient. This step never runs during play.

3

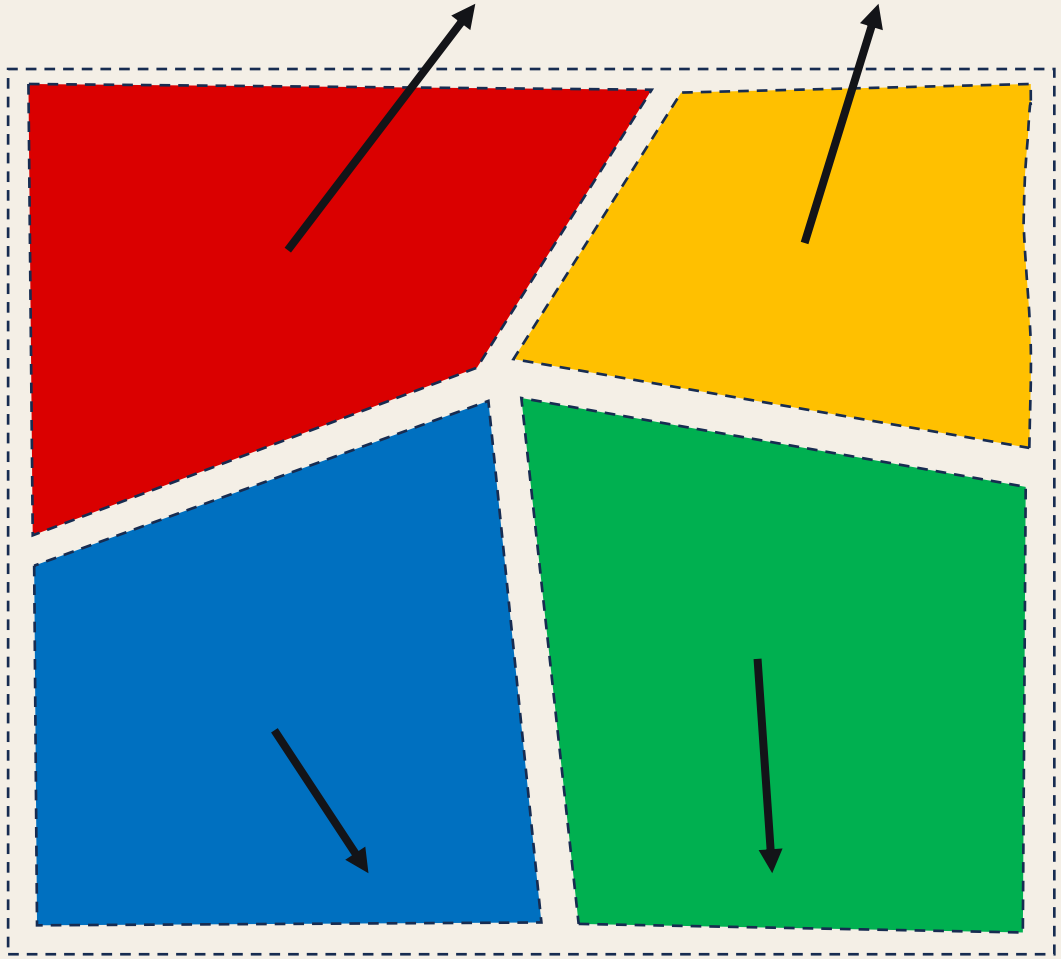
Train the experts

Disconnected L1 regions can share one expert, then each head is fine-tuned.

Knowledge-guided
Initialization and
Two-stage
Segregation for
Unsupervised
Neural
Experts

KITSUNE - Stitching

- You want B experts
- Partition the space to $B' = B\ k_{refinement}$ mini-clusters
- *Stitch* together the partitions with similar *average sample gradient*
- Accuracy vs B

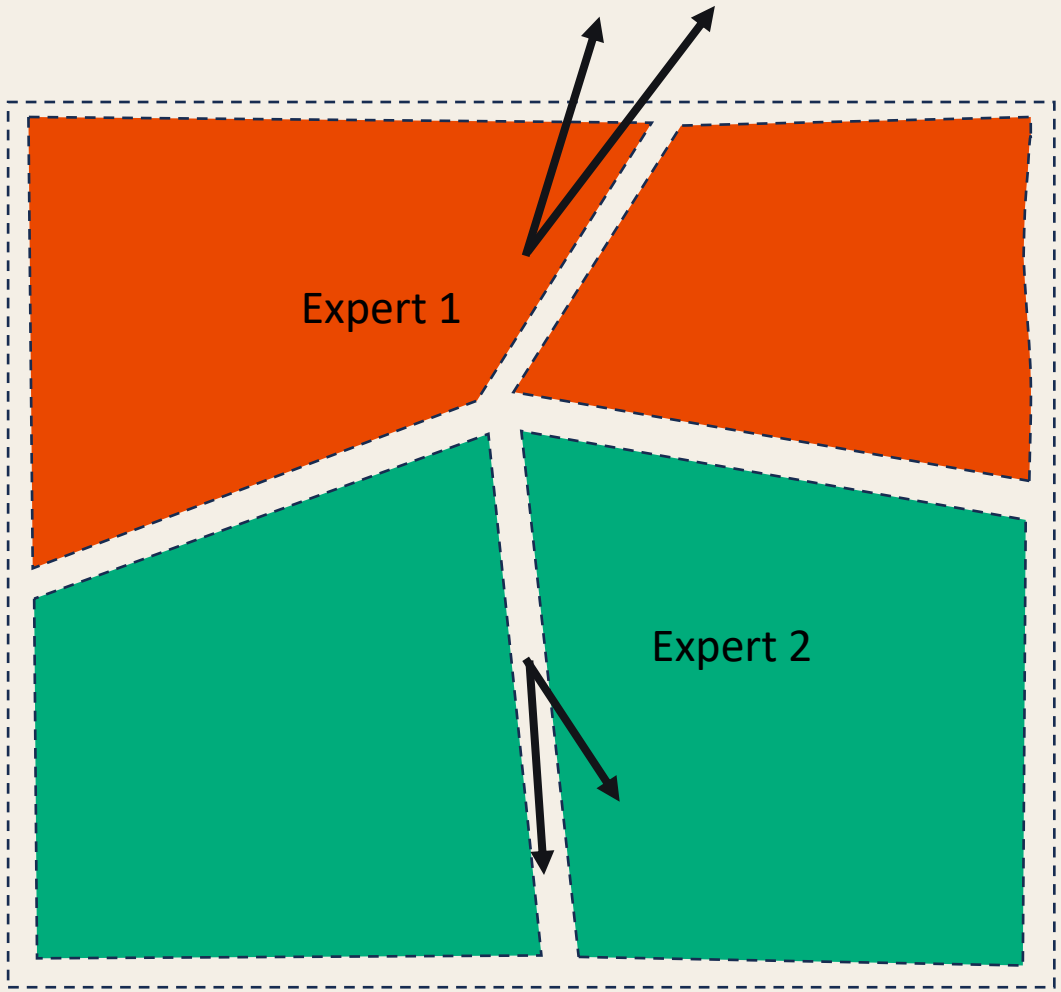


Mini-partitions of the L1 embedding space,
and their respective average sample gradients.

The activation space is first partitioned precisely, then the clusters with similar sample gradients are mearged

KITSUNE - Stitching

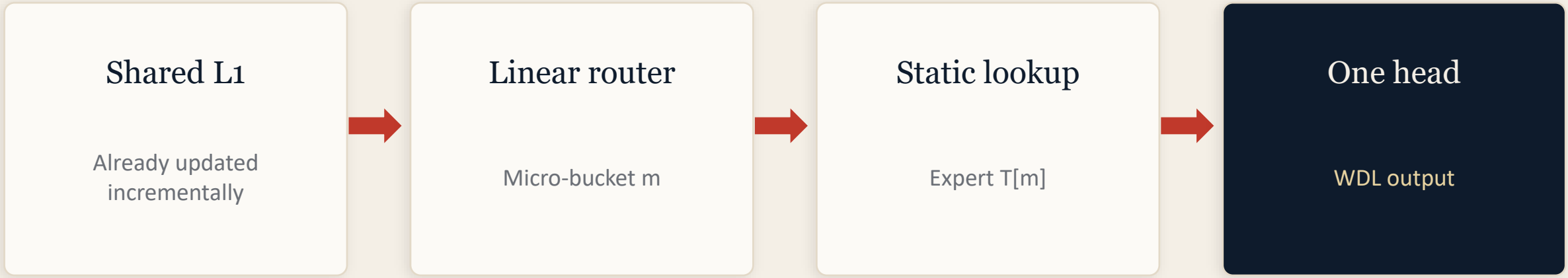
- You want B experts
- Partition the space to $B' = B\ k_{refinement}$ mini-clusters
- *Stitch* together the partitions with similar *average sample gradient*
- Accuracy vs B



Partitions of the L1 embedding space, and their respective average sample gradients.

The activation space is first partitioned precisely, then the clusters with similar sample gradients are mearged

Online inference needs no gradients



The online cost is a linear map plus a table lookup. Convex micro-buckets are what make that map enough.

The routed mixture on the weaker base is measured. It does not require a gradient at play time.

KITSUNE matches that L1 result...

0.65043

Routed mixture

K = 16, two micro-buckets per expert. 0.0033 below the base, and 0.0005 higher than the best L1 run.

0.60234

Best expert, own bucket

Factor 4. A 7.85% reduction against the base CE. Factor 2 is already at 0.60571.

Refinement factor at 16x

Still no significant improvement

At K = 16 the mixture rises to 0.65138 and the best expert to 0.60865.

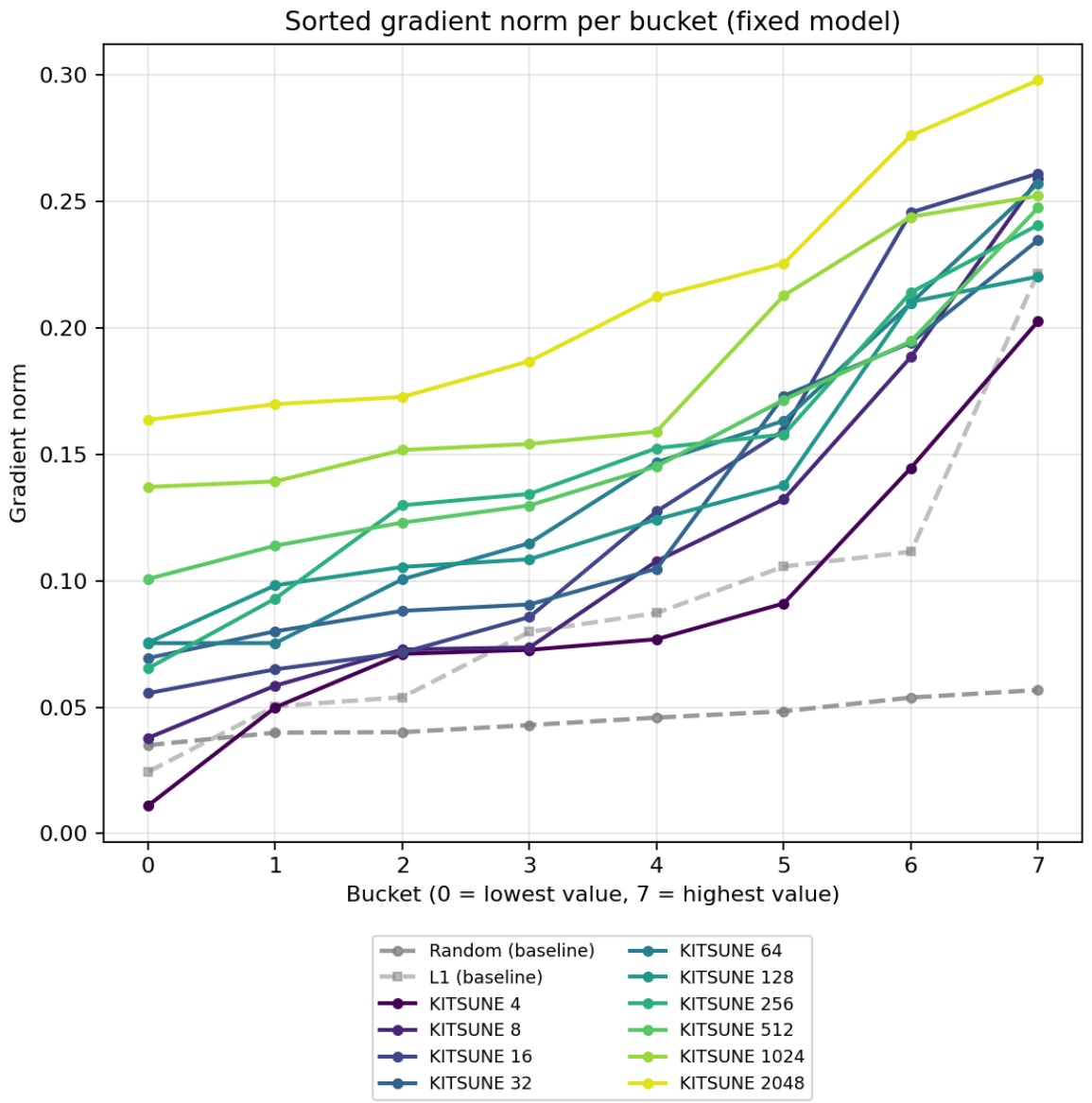
K	B′/K	MoE CE	Best expert	Gain
4	4	**0.65118**	**0.63726**	2.51%
4	8	**0.65137**	**0.64050**	2.01%
8	4	**0.65102**	**0.61748**	5.54%
8	8	**0.65093**	**0.61275**	6.26%
16	2	**0.65043**	**0.60571**	7.34%
16	4	**0.65061**	**0.60234**	7.85%
16	8	**0.65138**	**0.60865**	6.89%

Table 5.16. Expert gain uses the full-base CE, not a same-bucket comparison. Strong-base KITSUNE, Elo, and nodes per second are still open.

Some Last-minute Results!

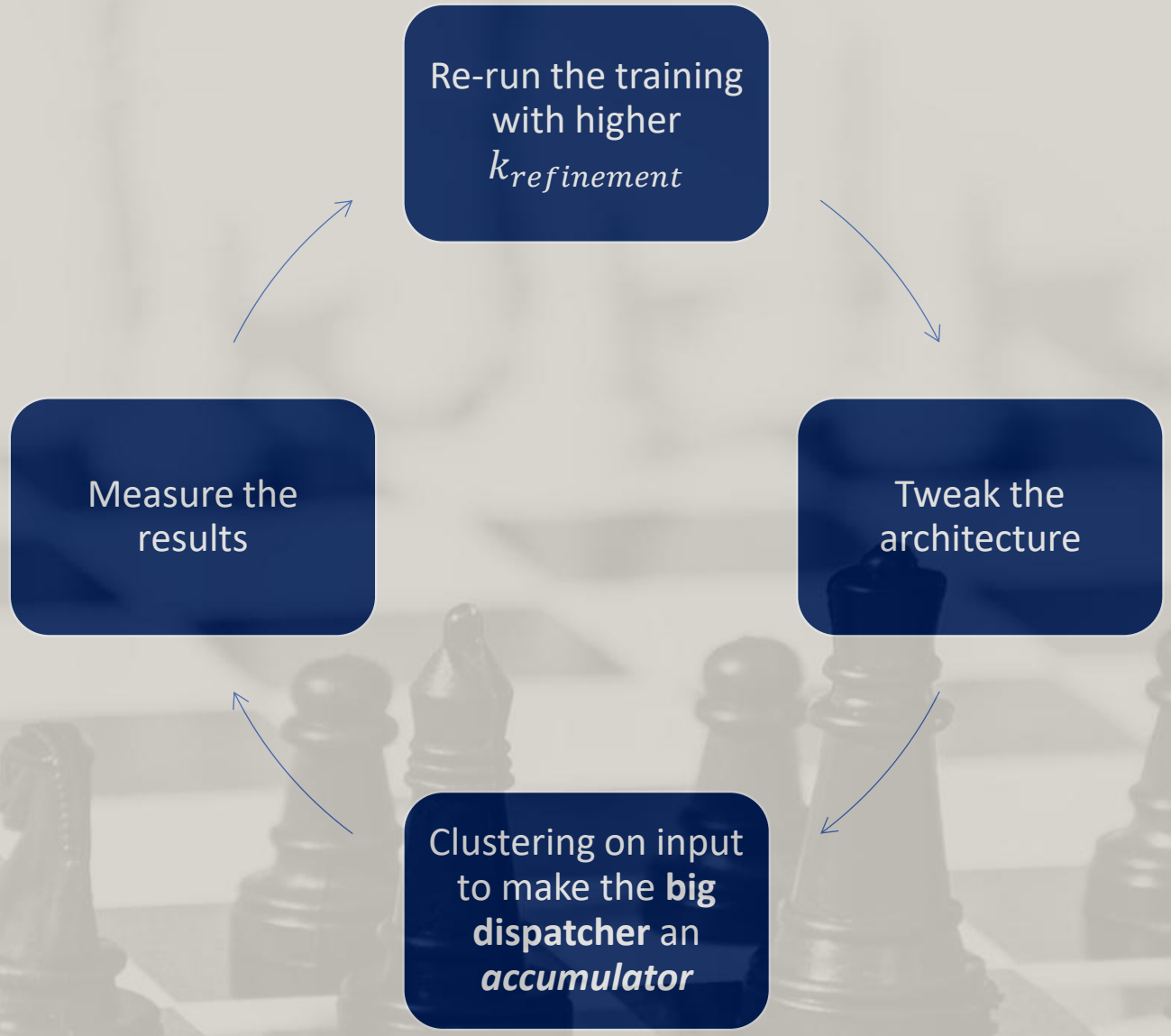
$k_{refinement}$ needs to be **way bigger**!

- More clusters = Less points in the cluster
  = More aligned sample gradients
- *Stitch* together **hundreds of partitions** with similar *average sample gradient*
- Accuracy on B' is not important
- Downside: the dispatcher gets really big!
  … but that is OK.



Each series represents gradient magnitudes of increasing strength for that algorithm. The more mini-clusters, the better!

Where do we go from here?



...

The improvement process for the following iterations of KITSUNE.

The take-home message

KITSUNE routes *robustly* with L1 clustering and a lookup.

Stitching seems to work **only** when the number of mini-clusters is very high.

We will leverage these all findings for the next iteration of KITSUNE.

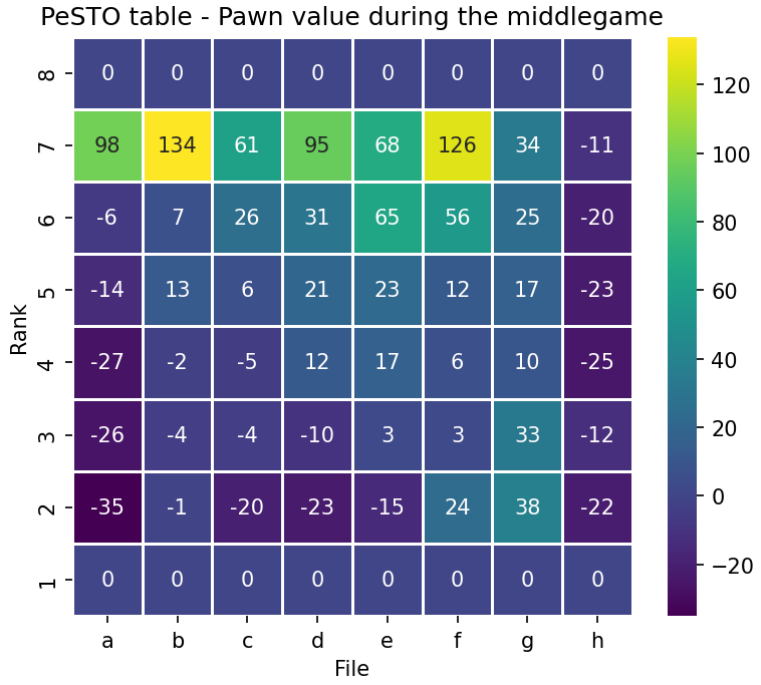
Thank you.

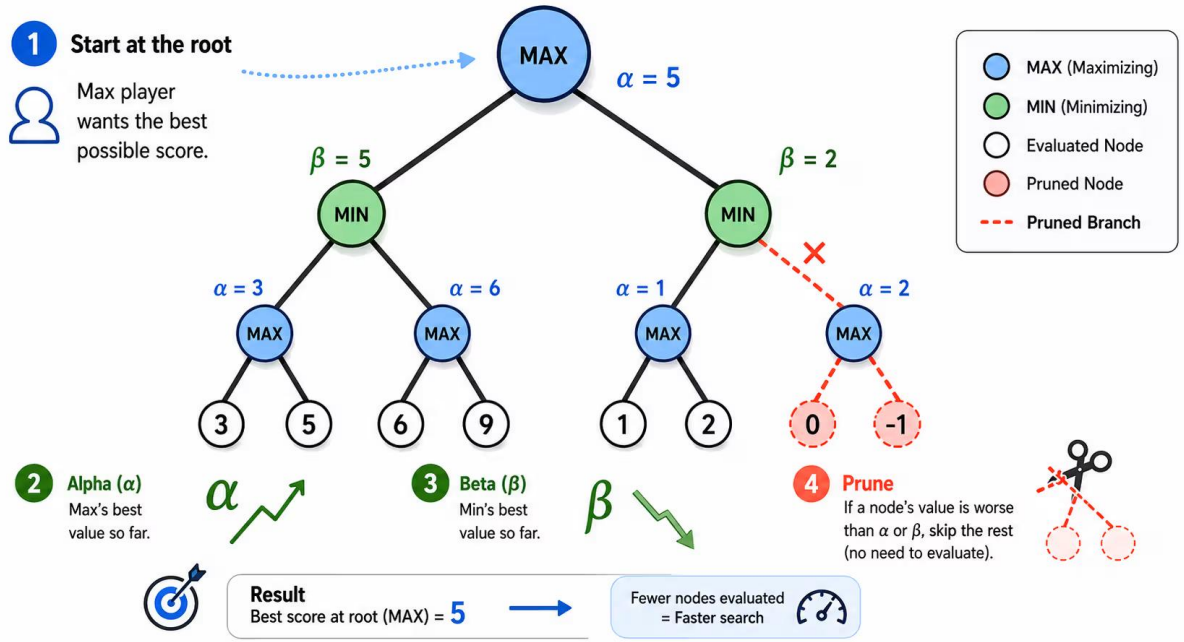
Questions?

Omar Cusma Fait · University of Trieste

Omar Cusma Fait · University of Trieste

Some more slides…





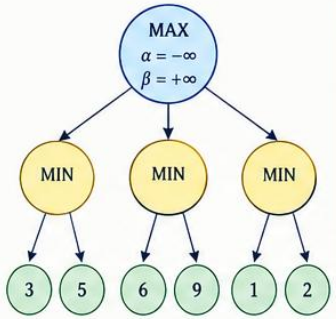
Alpha-Beta Tree Search

Explore promising moves, prune the rest.

Start at root → Use α (best for MAX) and β (best for MIN) → Search depth-first → Prune when $\alpha \geq \beta$ → Return best move

1 Initial state

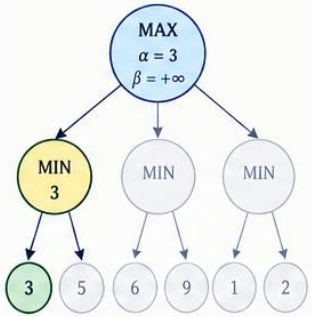
MAX plays at the root (blue), MIN plays at the next level (yellow).



All leaves are unvisited.

2 Explore left subtree

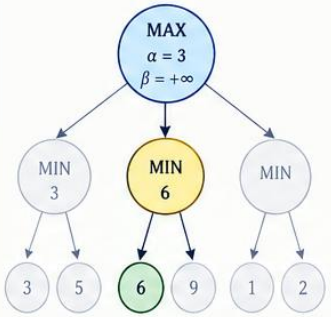
MAX considers the left child first. MIN returns 3 (min of 3 and 5).



α is updated to 3.
(Best value for MAX so far)

3 Explore middle subtree

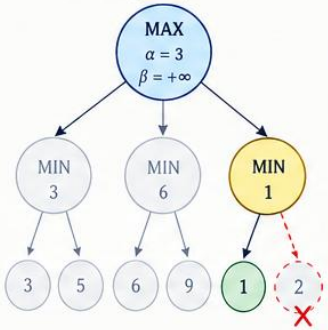
MIN node sees a value 6, so $\beta = 6$. Since $6 \leq \alpha$ (3) is false, we continue.



α remains 3.
(Best value for MAX so far)

4 Prune right subtree

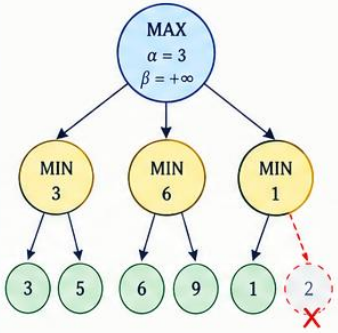
At the right MIN node, the first leaf is 1. Since $1 \leq \alpha$ (3), we prune the rest (2 is not explored).



β at the right MIN becomes 1. Since $1 \leq \alpha$ (3), prune remaining children of this node.

5 Result

The root chooses the maximum of the three explored values.



Best move value: 3
(from the left subtree)

α (alpha)

The best value for MAX found so far along the current path (lower bound). Initial value: $-\infty$

β (beta)

The best value for MIN found so far along the current path (upper bound). Initial value: $+\infty$

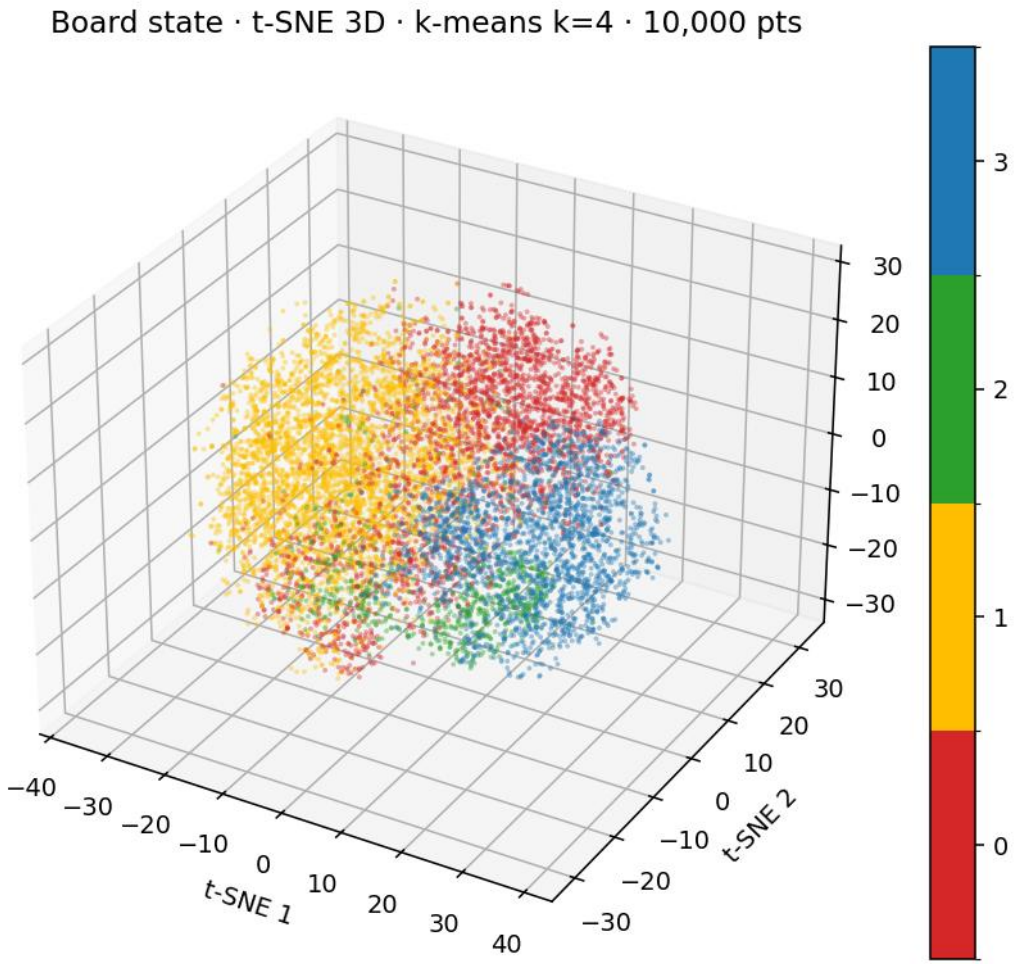
Pruning condition

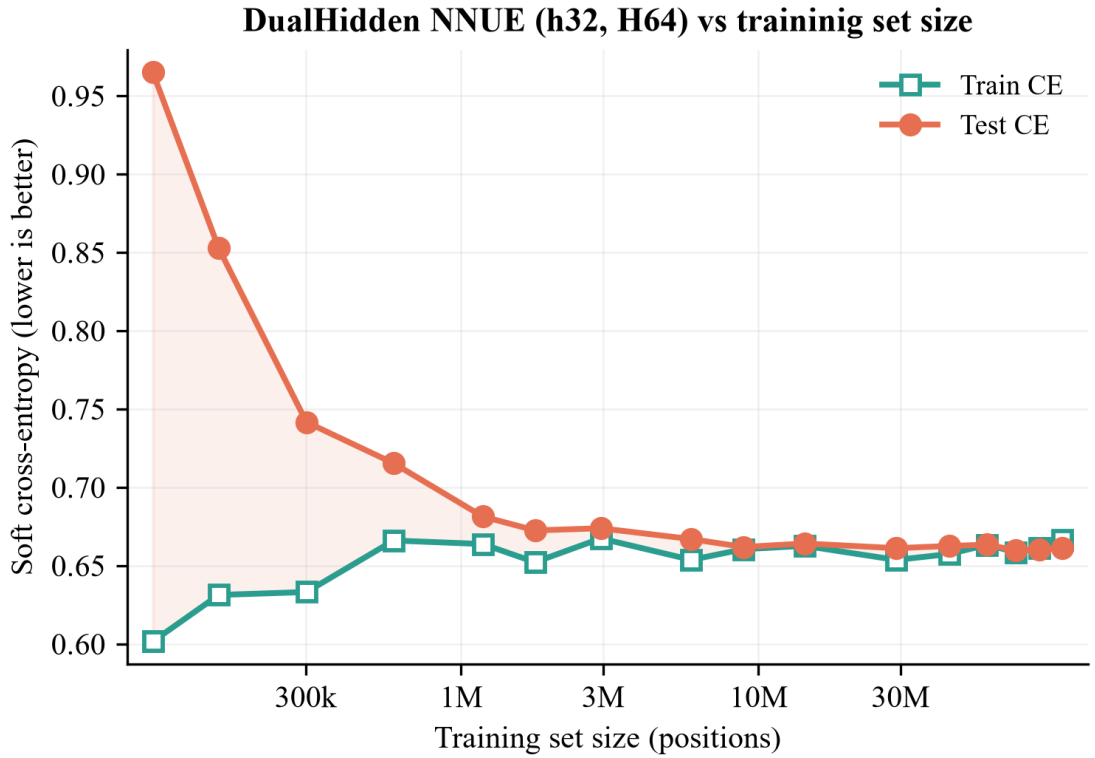
If $\alpha \geq \beta$, stop exploring this branch. The remaining moves in this subtree cannot affect the final decision.

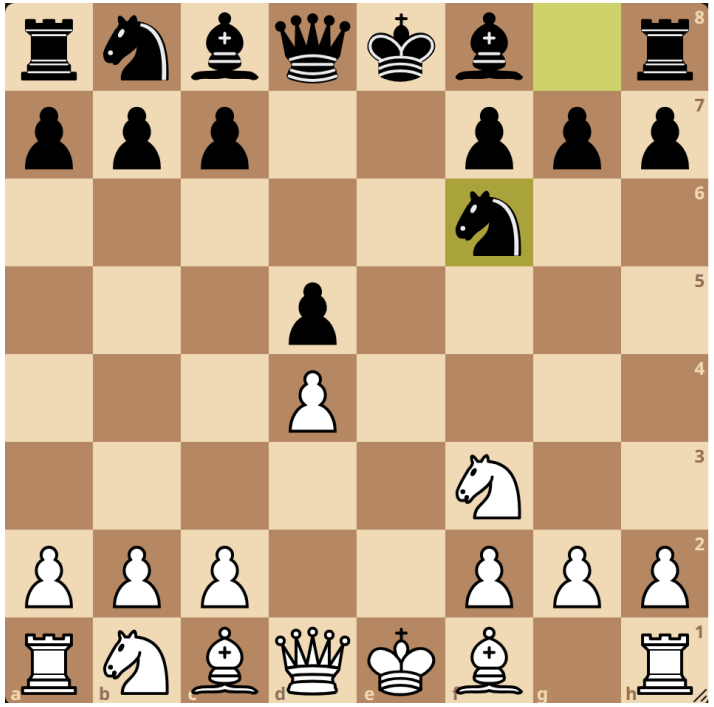
Pruned node

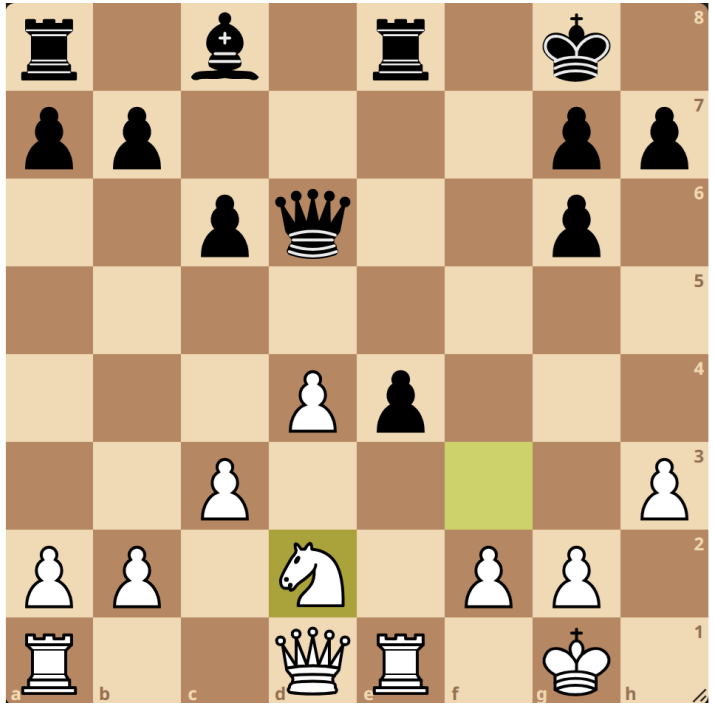
Not evaluated, because the branch cannot improve the result.

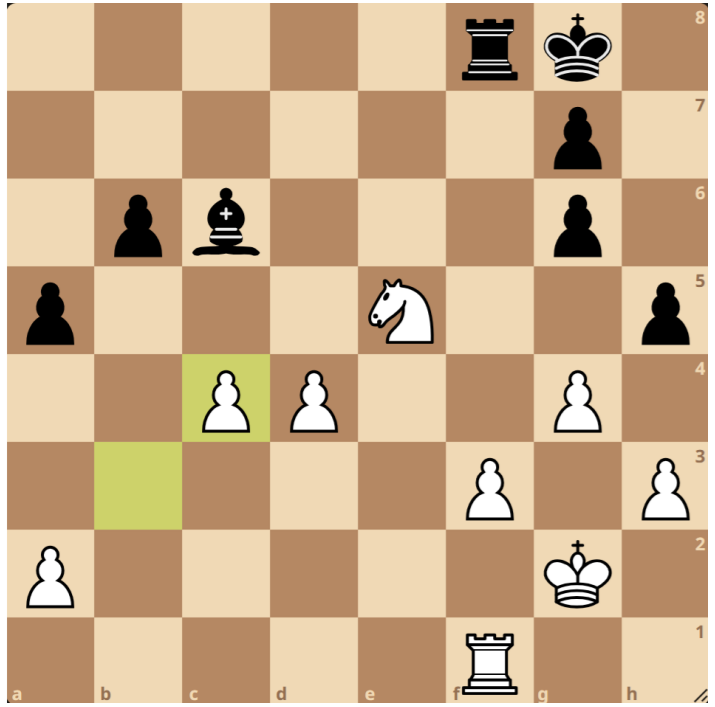
Paper/source	Contribution
Silver et al. (2018)	AlphaZero — superhuman performance with no human data
Nasu (2018)	**NNUE** — efficient and strong neural evaluation for shogi (base structure of out algorithms)
NNUE for **Stockfish**	Neural value evaluation for Stockfish
de Man (2016)	**Cfish** — C-based implementation of Stockfish for efficient microcontroller implementation (very useful, efficiently implements alpha-beta search)
Shazeer et al. (2017)	Sparsely-Gated Mixture-of-Experts
Fedus, Zoph & Shazeer (2021)	Switch Transformer—Simpified routing MoE with top-1 assignment
Kawata et al. (2025)	MoE and cluster latent structure – how much can a MoE exploit latent cluster structure of the data
Hu et al. (2021)	**LoRA** - Low-Rank Adaptation – training lower-rank matrices to fine-tune LLM (useful for big models, not necessary for us)
Li et al. (2025)	ELREA - Ensembles of Low-Rank Expert Adapters – clustering training instructions based on gradients
Sridharan et al. (2025)	GradientSpace - clustering of LoRA gradients for expert specialization
Zhang et al. (2025) — TAGCOS	Task-Agnostic Gradient Clustered Coreset Selection for Instruction Tuning Data


Board state · t-SNE 3D · k-means k=4 · 10,000 pts
t-SNE 1
t-SNE 2
−40
−30
−20
−10
0
10
20
30
40
3
2
1
0


DualHidden NNUE (h32, H64) vs traininig set size
Train CE
Test CE
Soft cross-entropy (lower is better)
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
300k
1M
3M
10M
30M
Training set size (positions)







THE DEPLOYMENT CONSTRAINT

Specialization must preserve search speed

120 MHz	192 KB	512 KB
CPU frequency	Internal RAM	Internal flash

Alpha–beta search calls the evaluator over and over. A slower network can cost a ply of search, which often matters more than a small gain in the value itself.

Target: Wio Terminal, ARM Cortex-M4.

The question

Can specialization improve the prediction inside this budget?

Hardware specifications, not measured deployment. Thesis §4.8.

OFFLINE STRUCTURE

Gradients form more separated clusters

Silhouette at K = 8, each score measured in its own representation.

0.191

Sample gradients

The most separated of the three signals.

0.003

L1 activations

Almost no silhouette. Easy to draw, little separation.

Separation is not the same as a better expert or a usable router. Table 5.2; gradient UMAP in Fig. 5.3.

Bucketing defines each expert's task

Bucket signal	Groups positions by	Available online?
Piece count	A hand-designed game-phase proxy	Yes — a fixed rule
L1 activations	Similar hidden representations	Yes — already computed
Sample gradients	Similar updates of the head	No — needs the teacher WDL
Jointly learned gate	An end-to-end routing objective	Yes — after training

L1 describes the position. Gradients describe how the head should change.

Thesis §3.3. Per-sample head gradient uses the teacher target, which is unknown at play time.

SAME GRADIENT-TRAINED EXPERTS, DIFFERENT SELECTION RULES

The oracle exposes assignment headroom

K = 16

0.6301

Actual dispatcher

What can run online.

K = 16

0.6280

Gradient-centroid proxy

Still needs the teacher target.

K = 16

0.6005

Minimum-loss oracle

Each position goes to its best expert.

Table 5.13. The single-head base is CE 0.6303. Centroid proxy and oracle both need the unavailable teacher target.

Longer fine-tuning helps a weaker base

BASE · W64 / H128

0.6537

Test CE of the single evaluation head.

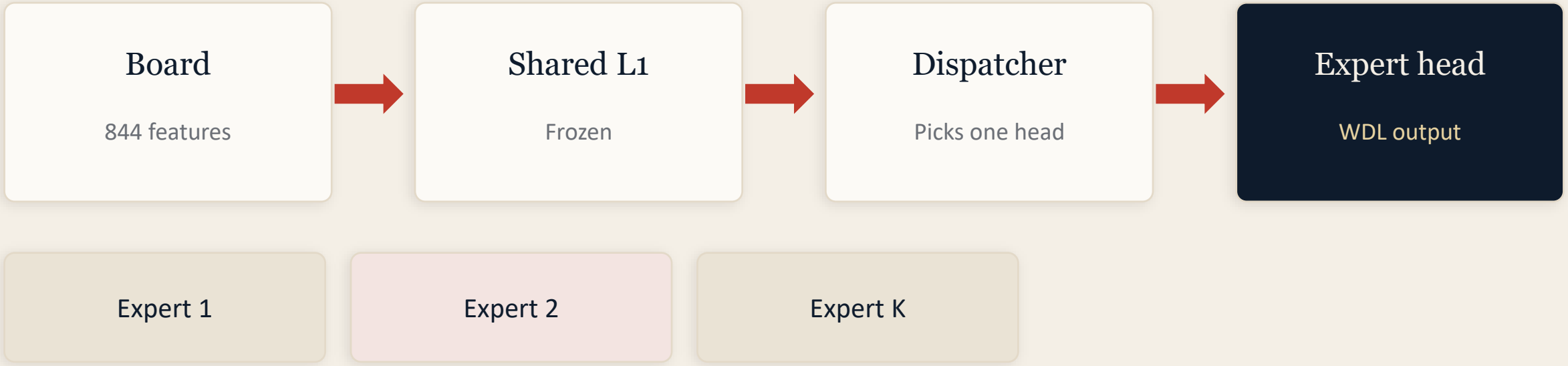
FOUR L1 EXPERTS

0.6499

Linear dispatcher, longer fine-tuning.
A reduction of 0.0038.

This is the weaker base, not the strong reference. §5.5.4, K = 4. The gain depends on the model and the training budget.

One accumulator, one selected expert



Fine-tuning stores several small heads. At inference, L1 activations select one of them. The others stay in memory and do not run.

Hard routing, thesis §§3.3–3.4. Extra experts still cost memory.



Nue



Kitsune